Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · www.solaress.org

SOLAR + BATTERY GUIDE

How long can batteries run your air conditioners?

A simple guide for 6 air cons on 4 × 16 kWh LiFePO4 batteries

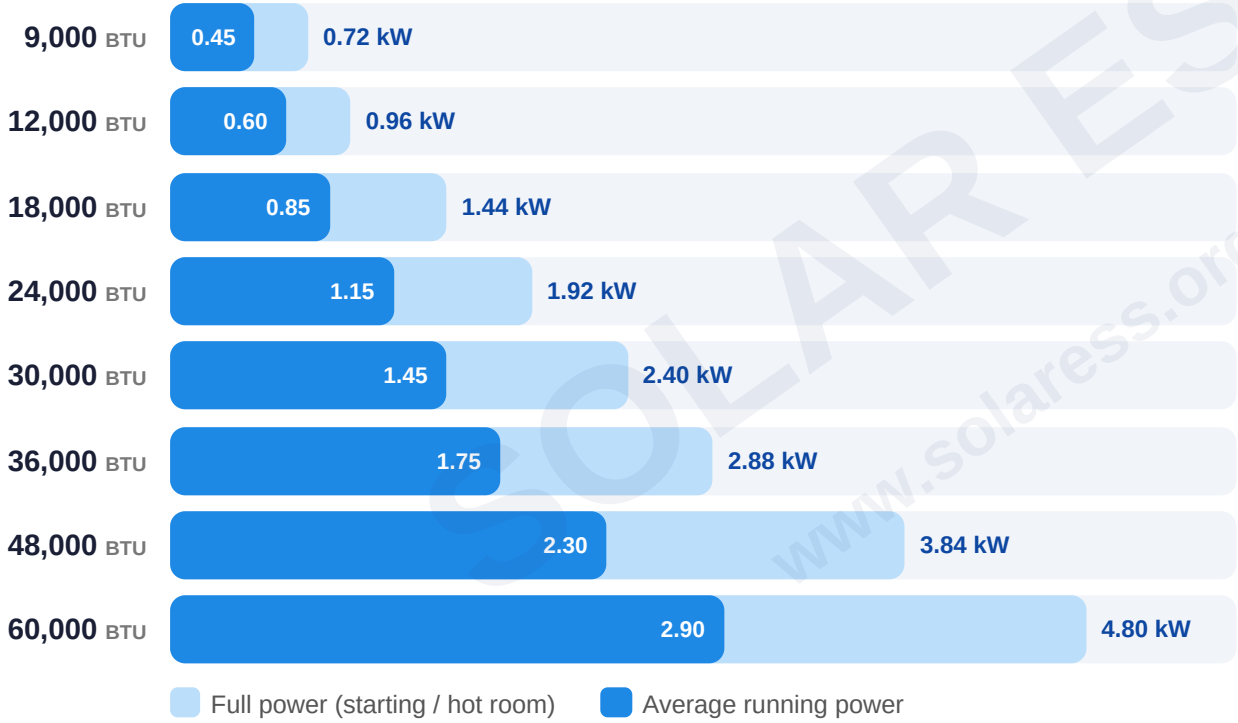
❄️ 6 air cons

🔋 4 × 16 kWh

⏱️ hours of cooling

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1 How much power does an air con use?



Figures for inverter air cons (EER ≈ 12.5). Non-inverter units use about 15% more.

Rule of thumb

12,000 BTU ≈ 1 kW

Power (kW) = BTU ÷ (EER × 1,000)

Inverter air cons slow down once the room is cool, so on average they use only **about 60%** of full power.

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2 How much energy do 4 batteries hold?



4 × 16 kWh
= 64 kWh total

×



90%
usable (DoD)

×



95%
inverter efficiency

=

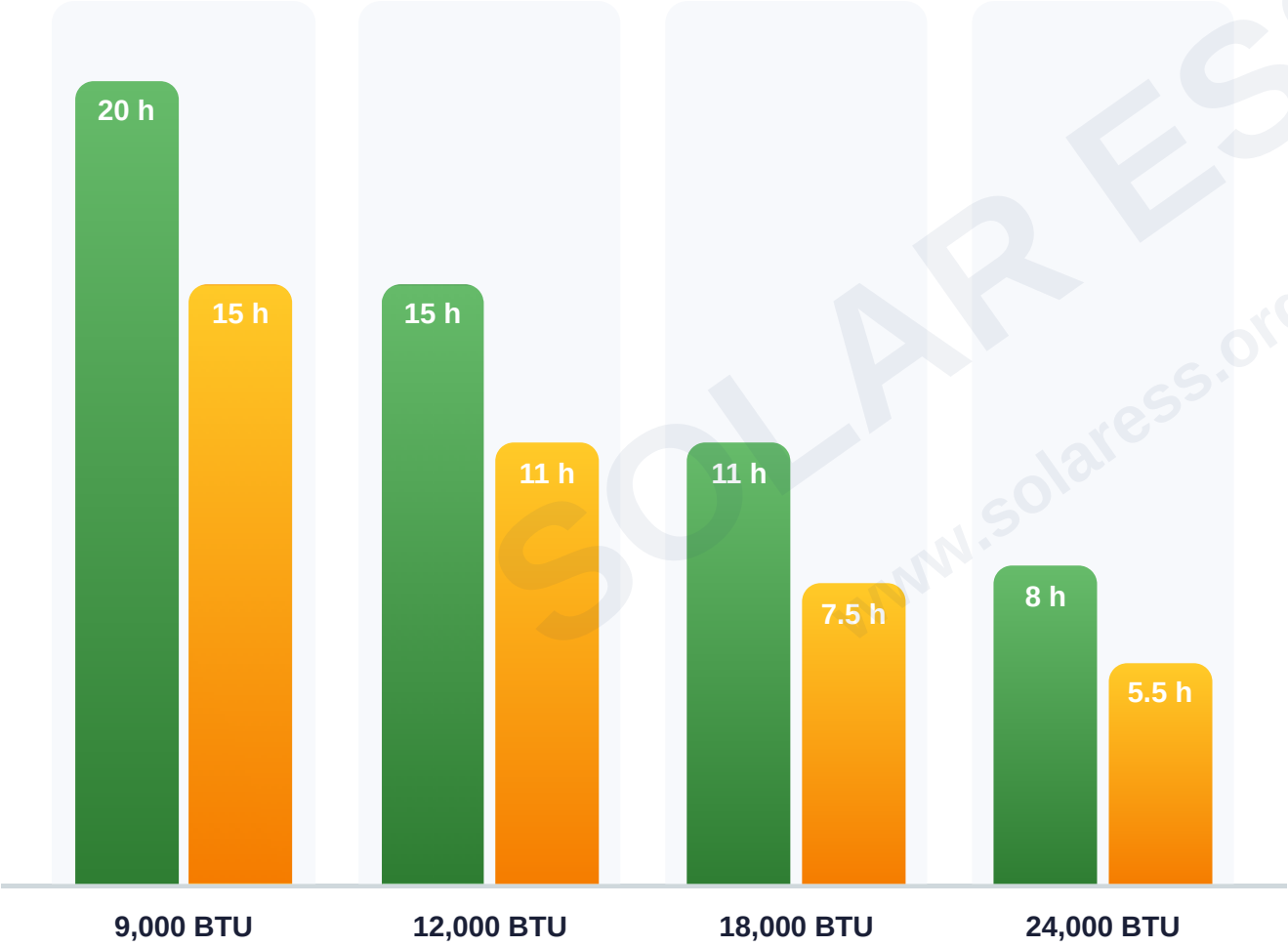
≈ 55 kWh
ready to use

💡 **55 kWh ≈ 55 units of electricity**, the same "units" shown on your PEA bill.

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3 How many hours for 6 air cons?



Hours = 55 kWh ÷ (6 × kW per air con)

Inverter air con Non-inverter air con

6 × 12,000 BTU inverter

≈ 15 hours

enough for the whole night

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4 Tips for the longest runtime



Set 25–26 °C

Every degree cooler adds roughly 6–8% more power use.



Choose inverter air cons

About 25–35% longer runtime than non-inverter units.



Start them one at a time

Air cons draw a surge when starting. Staggering protects the inverter.



Size the inverter too

6 × 12,000 BTU ≈ 6 kW → one 10 kW inverter is OK. 18,000–24,000 BTU needs two inverters.

☀️ Solar ESS designs the right battery and inverter size for your home or business · Tel 063 151 6694

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